

BEAVER MINES, AB, Canada

WMO: 713040

Lat: 49.4678N

Lon: 114.1750W

Elev: 4124

StdP: 12.63

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.7	-11.8	-24.2	1.7	-16.9	-18.1	2.4	-10.9	29.3	27.2	26.4	29.4	2.8	170	0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	83.9	64.5	80.7	62.5	77.5	60.8	72.1	79.1	69.1	75.4	65.6	71.7	8.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.8	128.1	77.4	66.5	114.0	72.9	63.0	100.7	69.6	38.7	78.2	35.9	75.7	32.7	72.0	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	21.2	18.6	-27.4	89.8	9.4	4.2	-34.2	92.8	-39.7	95.2	-45.0	97.6	-51.9	100.6
			-26.4	67.7	9.5	7.6	-33.3	73.2	-38.9	77.6	-44.2	81.9	-51.2	87.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.0	22.5	17.7	28.8	40.0	49.7	57.3	63.1	61.4	53.5	42.0	32.4	22.3	(6)
(7)		DBStd	18.85	14.98	15.72	13.60	8.80	8.09	7.12	5.72	6.33	8.84	9.47	11.46	14.23	(7)
(8)		HDD50	4572	852	905	658	310	106	11	1	2	64	276	529	858	(8)
(9)		HDD65	8859	1316	1325	1122	750	478	254	103	146	353	712	977	1323	(9)
(10)		CDD50	1305	1	0	1	11	97	230	408	356	168	30	3	0	(10)
(11)		CDD65	113	0	0	0	0	3	23	45	35	7	0	0	0	(11)
(12)		CDH74	1584	0	0	0	1	51	250	685	512	77	8	0	0	(12)
(13)		CDH80	334	0	0	0	0	9	52	150	114	9	0	0	0	(13)
(14)	Wind	WSAvg	7.6	8.7	8.1	8.4	8.4	7.1	6.9	5.8	5.5	6.3	8.2	8.6	8.9	(14)
(15)	Precipitation	PrecAvg	23.0	1.5	1.2	1.4	2.2	3.0	3.2	1.8	2.1	2.4	1.2	1.5	1.5	(15)
(16)		PrecMax	33.1	3.9	3.8	2.7	5.6	7.1	11.5	6.5	5.7	6.9	3.2	3.5	3.0	(16)
(17)		PrecMin	13.0	0.3	0.2	0.1	0.3	0.7	0.4	0.2	0.1	0.1	0.0	0.0	0.2	(17)
(18)		PrecStd	5.4	0.9	0.7	0.7	1.1	1.6	2.5	1.7	1.5	1.7	0.9	0.8	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	50.4	57.4	71.7	80.3	85.2	87.2	87.8	80.8	73.3	61.0	49.7	(19)
(20)			MCWB	40.1	39.3	44.0	50.7	67.7	69.4	64.5	65.1	59.0	63.7	47.9	40.0	(20)
(21)		2%	DB	45.6	46.3	52.4	64.7	73.6	80.7	84.1	82.7	75.7	63.9	55.2	44.8	(21)
(22)			MCWB	37.7	38.2	40.3	47.5	55.0	65.9	64.3	61.4	58.9	51.0	46.2	36.8	(22)
(23)		5%	DB	42.6	43.3	48.8	58.9	69.6	76.4	80.8	79.8	71.9	59.7	50.1	41.5	(23)
(24)			MCWB	34.5	35.5	39.5	44.9	53.3	62.4	62.8	60.7	57.0	48.6	43.7	35.2	(24)
(25)		10%	DB	40.5	39.3	45.8	53.8	65.7	72.0	77.5	76.5	68.5	55.8	46.1	38.8	(25)
(26)			MCWB	33.1	32.1	37.2	42.2	51.6	59.8	61.1	58.8	55.7	46.2	40.1	32.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.6	40.5	48.2	52.8	67.5	74.4	75.3	75.3	67.9	64.0	54.7	42.8	(27)
(28)			MCDWB	44.9	47.3	52.8	66.0	75.4	83.0	81.6	81.5	76.1	70.9	57.2	45.5	(28)
(29)		2%	WB	39.0	38.6	42.4	49.3	60.6	70.9	72.5	69.8	64.7	55.6	46.6	38.7	(29)
(30)			MCDWB	44.9	45.7	50.2	59.5	69.6	77.4	79.1	77.0	70.1	60.2	53.2	43.5	(30)
(31)		5%	WB	35.9	35.9	39.6	46.2	55.7	66.4	69.8	66.2	62.0	51.2	43.8	35.3	(31)
(32)			MCDWB	41.1	42.3	47.9	57.7	64.4	71.8	75.8	72.2	66.5	56.4	49.3	40.6	(32)
(33)		10%	WB	33.4	32.6	37.4	43.4	52.4	61.1	65.7	62.7	59.0	47.4	40.5	33.1	(33)
(34)			MCDWB	39.3	38.8	45.2	52.9	63.1	69.8	71.4	69.8	63.5	54.8	45.6	37.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.2	15.6	16.0	17.4	19.2	19.1	23.6	23.7	19.0	15.3	13.6	13.1	(35)
(36)			MCDBR	13.3	12.1	18.4	29.2	27.9	25.4	29.3	29.4	26.6	21.4	19.3	12.4	(36)
(37)			MCWBR	9.3	8.6	11.0	15.1	13.7	12.1	13.3	12.6	13.2	12.3	13.4	9.0	(37)
(38)		5% WB	MCDWB	13.4	15.1	17.8	27.3	23.2	20.4	21.8	21.9	20.5	20.6	16.9	12.0	(38)
(39)			MCWBR	11.4	12.4	12.1	15.7	13.4	11.9	13.5	12.3	13.5	14.5	12.6	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.240	0.250	0.280	0.340	0.344	0.337	0.334	0.348	0.314	0.281	0.258	0.230	(40)
(41)		taud		2.391	2.375	2.393	2.269	2.372	2.434	2.456	2.412	2.487	2.562	2.501	2.414	(41)
(42)		Ebn at Noon		268	291	298	287	290	292	291	282	283	277	259	257	(42)
(43)		Edh at Noon		20	26	30	39	36	35	33	33	28	22	18	16	(43)
(44)	All-Sky Solar Radiation	RadAvg		362	673	1024	1456	1737	1898	2180	1772	1240	749	390	285	(44)
(45)		RadStd		34	54	93	113	147	150	154	122	121	79	36	27	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page